



A 960 μ W 10-bit 70-MS/s SAR ADC with an energy-efficient capacitor-switching scheme

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ARTICLE INFO

Article history:

Received 10 March 2013

Received in revised form

10 August 2013

Accepted 19 August 2013

Available online 7 September 2013

Keywords:

Successive approximation register (SAR)
analog-to-digital converter (ADC)

Split-capacitor V_{cm} -based cap-switching
scheme

Comparator DC offset correction

Asynchronous self-timing

ABSTRACT

This paper presents a low-power 10-bit 70-MS/s successive approximation register (SAR) analog-to-digital converter (ADC) using a novel energy-efficient capacitor-switching scheme. Compared to the conventional scheme, the proposed split-capacitor V_{cm} -based capacitor-switching scheme can reduce the capacitor-switching energy by about 92% with better monotonicity. Meanwhile, full-custom SAR logic and registers, variable-delay self-timing cell and dynamic comparator with proposed two-segment DC offset correction scheme are also implemented to improve the conversion speed and accuracy requirements. The prototype was fabricated in 65-nm 1P9M CMOS technology. Measurement results show a peak signal-to-noise-and-distortion ratio (SNDR) of 53.2 dB, while consuming 960 μ W from 1.2 V supply voltage. The figure of merit (FoM) is 36.8 fJ/conversion-step and the total active area is only $220 \times 220 \mu\text{m}^2$.

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1. Introduction

With the popularity of mobile applications and “green design”, the low-power design has been more and more concerned [1]. Due to the low-power advantage and the compatibility with process scaling, successive approximation register (SAR) analog-to-digital converters (ADCs) have been widely used in mobile systems and become a potential competitor in moderate resolution, medium-or-high-speed ADCs [1–5].

In a SAR ADC, the primary sources of energy consumption are the SAR control logic, low-noise comparator, and capacitive digital-to-analog converter (DAC) [2]. The advancement of technology reduces the digital energy consumption while the energy consumption of comparator and capacitive DAC is limited by the noise and matching requirements. In the capacitive DAC, particularly, the switching energy can be greatly reduced by modifying the capacitor-array structure and the scheme the capacitors are switched. Recently, several schemes have been proposed, including split capacitor [3], set-and-down [2] and V_{cm} -based [4] schemes, which achieve 37%, 81%, and 87.5% energy saving, respectively, compared with that of the conventional.

In the V_{cm} -based scheme [4], two identical capacitor arrays are switched differentially relative to the common-mode voltage (V_{cm}). The binary-weighted capacitors in each array share a common top plate for sampling and comparison, and their individual bottom plates for transition during the subsequent bit steps. During the sampling

phase, differential input signals are stored on both common top plates with their individual bottom plates connected to V_{cm} . During each conversion cycle, the bottom plates of the capacitors are switched in descending order, from V_{cm} to V_{ref} (termed “up” transition) or ground (termed “down” transition) depending on each bit decision. Although it saves much energy due to V_{cm} , there is still unnecessary energy drawn from the reference voltage V_{ref} when the capacitors in the same array perform both kinds of transitions during one conversion cycle.

In order to eliminate such unnecessary energy waste, this work proposes split-capacitor V_{cm} -based capacitor-switching scheme [6]. It applies the split capacitor technique [3] to the V_{cm} -based scheme [4], avoiding “up” and “down” transitions occurring in the same array during each conversion cycle, and therefore further saving energy. The proposed scheme can reduce the switching energy by about 37.5%, compared with the V_{cm} -based, and at the same time have a better static performance of linearity with little expense of increased number of switches.

The rest of the paper is organized as follows. Section 2 describes the architecture of this SAR ADC and Section 3 details the principle of the proposed capacitor-switching scheme. Section 4 presents the circuit implementation of the key building blocks, with emphasis on the parasitic effects on the DAC implementation. Then, Section 5 shows the measurement results, and the conclusions are drawn in Section 6.

2. ADC architecture

Fig. 1 shows the block diagram of this SAR ADC. It consists of differential capacitor DACs with front-end bootstrapped switches,

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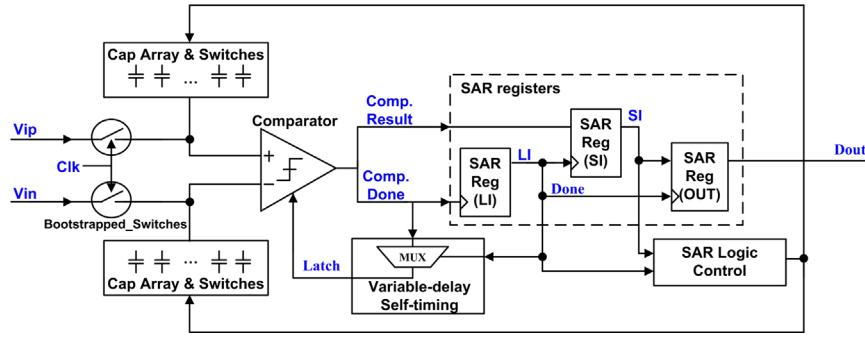
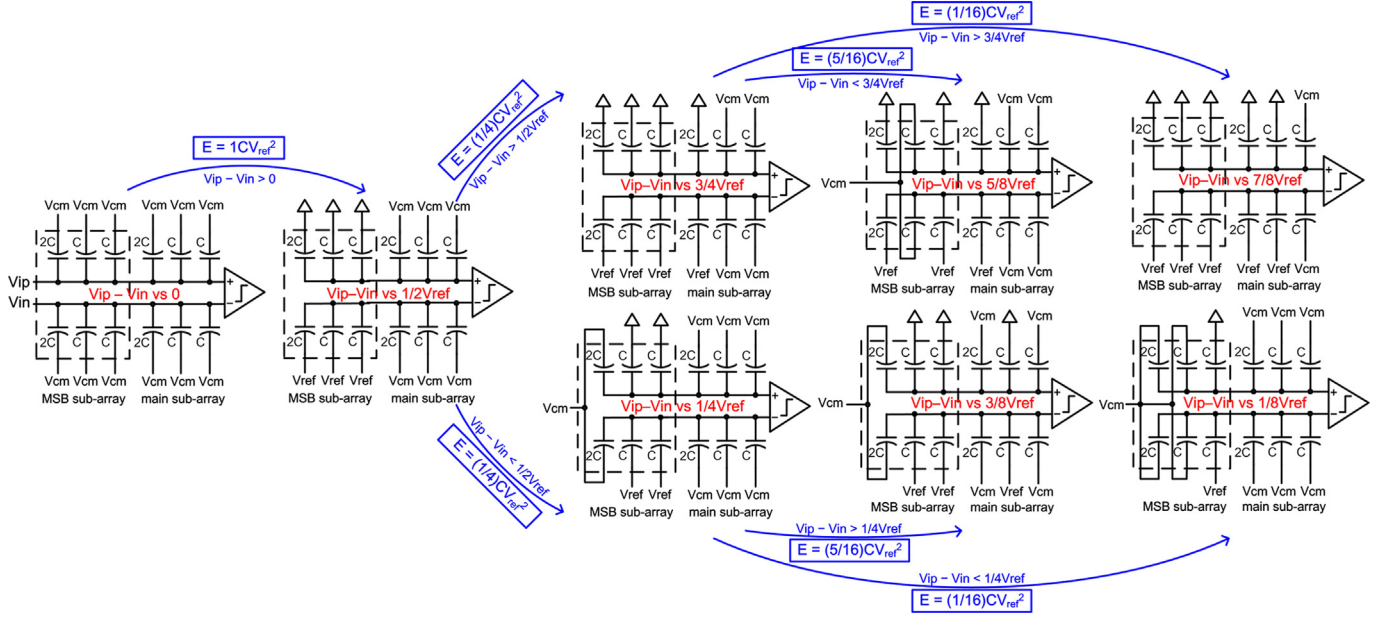


Fig. 1. The proposed SAR ADC architecture.

Fig. 2. Split-capacitor V_{cm} -based capacitor-switching scheme and energy consumption at each bit steps for 4-bit SAR ADC.

dynamic comparator containing proposed DC offset correction cell, variable-delay self-timing cell, and SAR registers with full-custom SAR logic. Differential bootstrapped switches and capacitor arrays operate as a sample-and-hold circuit. The following comparator then starts a first comparison. Based on the preview result, SAR logic determines the direction of the switches in the capacitor arrays and generates a new pair of differential voltages at the capacitors' top plates to prepare for the next comparison. The variable-delay self-timing cell controls the delay time between adjacent comparisons, giving the top-plate voltages enough time to setup. The results of ten comparisons are stored in SAR registers, which consist of several groups of registers with "Reg LIs" to flag on-going bit step, "Reg SIs" to save the result from the comparator in each bit step and "Reg OUTs" to export the final 10-bit results when one conversion cycle accomplishes.

3. The proposed split-capacitor V_{cm} -based capacitor-switching scheme

In the proposed split-capacitor V_{cm} -based capacitor-switching scheme [6], the Most-Significant Bit (MSB) capacitor $2^{n-2}C$ (C is the unit capacitor) is split into a copy of the rest of the main array, and thus, the array is divided into an MSB sub-array and a main sub-array. After the first comparison, the differential MSB sub-arrays are decided to "up" transitions or "down" transitions, respectively. During the subsequent bit steps, transitions depend

on the MSB sub-array transition in the same array. In the array with the "up" transition MSB sub-array, the capacitors in the main sub-array only allow "up" transitions, while for any required "down" transition, the corresponding capacitor in the MSB sub-array switches from V_{ref} to V_{cm} (termed "anti-up" transition). Similarly, in the array with the "down" transition MSB sub-array, the capacitors in the main sub-array only allow "down" transitions, while for any required "up" transition, the corresponding capacitor in the MSB sub-array switches from GND to V_{cm} (termed "anti-down" transition). In order to minimize switching energy, V_{cm} is equal to $V_{ref}/2$ to guarantee charge compensation at the V_{cm} terminal, although the departure from $V_{ref}/2$ does not affect the SAR ADC decision results.

Using a 4-bit SAR ADC as an example [6], Fig. 2 depicts the capacitor array structure and the switching procedure when MSB=1 in the proposed scheme. The MSB capacitor $4C$ is split into $2C$, $1C$, and $1C$ in parallel. After the sampling phase, the input differential signal is directly compared with 0 and the 1st comparing result (MSB) determines the MSB sub-array transition. The subsequent n th transition as well as its switching energy is identical to V_{cm} -based scheme if the following n -bit comparing results are the same as the MSB result. Otherwise, the transition becomes different. Taking MSB=1, i.e., $V_{ip}-V_{in} > 0$, in Fig. 2 for example, if the 2nd comparison is decided to be 0 ($V_{ip}-V_{in} < 1/2V_{ref}$), the two $2C$'s in the differential MSB sub-arrays short their bottom plates to V_{cm} , which are "anti-up" and "anti-down" transition, respectively, rather than the corresponding $2C$'s in the

main sub-array performs “up” or “down” transition as the V_{cm} -based scheme. In this way, the energy drawn from V_{ref} is $(1/4)CV_{ref}^2$, much less than $(5/4)CV_{ref}^2$ in the V_{cm} -based scheme. Subsequently, depending on the 2nd decision, the input differential signal is compared with $(1/4)V_{ref}$ or $(3/4)V_{ref}$. If the 3rd comparison is decided to be 0, the two C 's in the MSB sub-arrays short their bottom plates to V_{cm} and the energy drawn from V_{ref} is $(1/16)CV_{ref}^2$ and $(5/16)CV_{ref}^2$, respectively, much less than $(9/16)CV_{ref}^2$ and $(13/16)CV_{ref}^2$ in the V_{cm} -based scheme. No matter what type of transition occurs, no energy is drawn from V_{cm} due to the charge compensation effect.

Fig. 3 compares the proposed scheme with the V_{cm} -based scheme [4] in terms of switching energy consumed to obtain each code in a 10-bit SAR ADC. As shown in Fig. 3, in the proposed scheme, the switching energy versus output code has an “M” shape, symmetrical with respect to the middle code 512.

Assuming that the input signal uniformly distributed in the range of V_{ref} , the average switching energy of the proposed scheme for an n -bit SAR ADC can be expressed as

$$E_{average} = \left[2^{n-4} + \sum_{i=2}^{n-1} 2^{n-2-2i} (2^{i-1} - 1) \right] CV_{ref}^2 \quad (n \geq 3) \quad (1)$$

which, as well as the numbers of switches and capacitors required, is compared with the features of other four schemes in Table 1. Due to avoiding “up” and “down” transitions occurring in the same array during one conversion cycle, the proposed scheme has the best power efficiency. Despite some additional expense of increased number of switches, the SAR logic will yield little increase in digital power consumption. At the same time, applying the split scheme in the capacitor-array structure, a SAR ADC has an additional advantage of obtaining similar integral nonlinearity

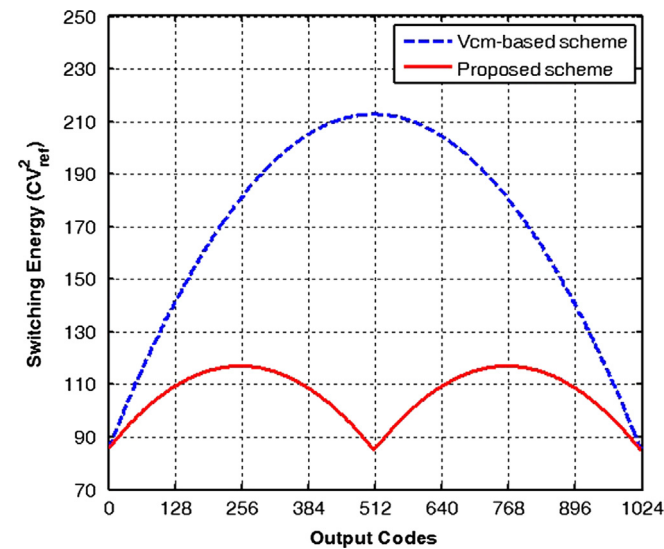


Fig. 3. Switching energy versus output code for the V_{cm} -based and proposed schemes for 10b SAR ADC.

Table 1
Comparison of switching procedures for n -bit SAR ADC.

Cap-switching schemes	Conventional	Split cap. [3]	Set-and-down [2]	V_{cm} -based [4]	Proposed
Normalized energy (%)	100	62.50	18.75	12.50	7.81
No. of switches	$6n+6$	$12n-2$	$4n-2$	$6n-4$	$12n-12$
No. of binary capacitors	$2n+2$	$4n$	$2n$	$2n$	$4n-4$
No. of unit capacitors	2^n	2^n	2^{n-1}	2^{n-1}	2^{n-1}

(INL) and better differential nonlinearity (DNL) performances due to better monotonicity characteristic [3]. By simulation, Fig. 4 compares the DNL monotonicity performance in 10,000 random iterations between the traditional V_{cm} -based scheme and the proposed split-capacitor V_{cm} -based scheme in 10b ADC model with the same unit capacitor. The capacitor relative mismatch parameter is 0.29%/1sigma. It shows that the proposed split-capacitor V_{cm} -based has a better DNL monotonicity characteristic.

Fig. 5 compares the average switching energy as the function of resolution between the two schemes and illustrates the saved average energy normalized to that in the V_{cm} -based scheme [4]. The percentage of average energy saved by the proposed scheme increases with resolution and converges towards 37.5%. It is worth noticing that such resolution dependence becomes weaker when the resolution is greater than 5 bits.

4. ADC circuit implementation

4.1. Capacitive DAC structure

With a view to low power, the capacitive DAC is designed by combining the proposed split-capacitor V_{cm} -based scheme with the bridge capacitor approach. The former reduces the switching energy as addressed above, and the latter reduced the total capacitance. As shown in Fig. 5, the energy efficiency of the proposed capacitor-switching scheme on number bit of resolution becomes weaker when the resolution is greater than 5 bits. The bridge capacitor divides one array into 4-bit LSBs array and 5-bit MSBs array, and the split-capacitor V_{cm} -based scheme is applied to the 5-bit MSBs array, shown in Fig. 6. Such segmentation is a trade-off between reduction of total capacitance and switching energy. Because of the limits on technology matching and thermal noise, the unit capacitor has been chosen to be 15 fF in this design.

One common potential issue of the bridge capacitor-array structure is the parasitic capacitance at the MSBs-array top plate and LSBs-array top plate [7,8]. Fig. 7 shows a general structure of a $(M+L+1)$ -bit SAR ADC bridge capacitor-array. If only a single channel is used in the ADC design, the problem of MSBs-array top-plate parasitic capacitance (C_{PM}) will be regarded as a gain error, which does not have any influence on the ADC's static and dynamic performance. However, the LSBs-array top-plate parasitic capacitance (C_{PL}) and the parasitic coupling capacitance between LSB_TOP and MSB_TOP (C_{PB}) will give rise to departure from the desired binary weighted relationship across the bridge, and result in poor linearity [7], which lead to a regular segmental curve in ADC's INL/DNL performance. A capacitor weight is the voltage contribution to the MSB_TOP node as its bottom plate jumps by the same voltage. Therefore, the weight of 1C in LSBs-array and the weight of 1C in MSBs-array can be expressed as

$$W(1C \text{ in LSB}) = \frac{C}{(2^L-1)C + C_{PL} + C_{eq,MSB}} \cdot \frac{C + C_{PB}}{C + C_{PB} + (2^M-1)C + C_{PM}} \\ = \frac{C(C + C_{PB})}{(2^{M+L}-1)C^2 + [2^L C_{PL} + (2^M + 2^L - 2)C_{PB} + 2^L C_{PM}]C + C_{PL}C_{PB} + C_{PL}C_{PM} + C_{PB}C_{PM}} \quad (2)$$

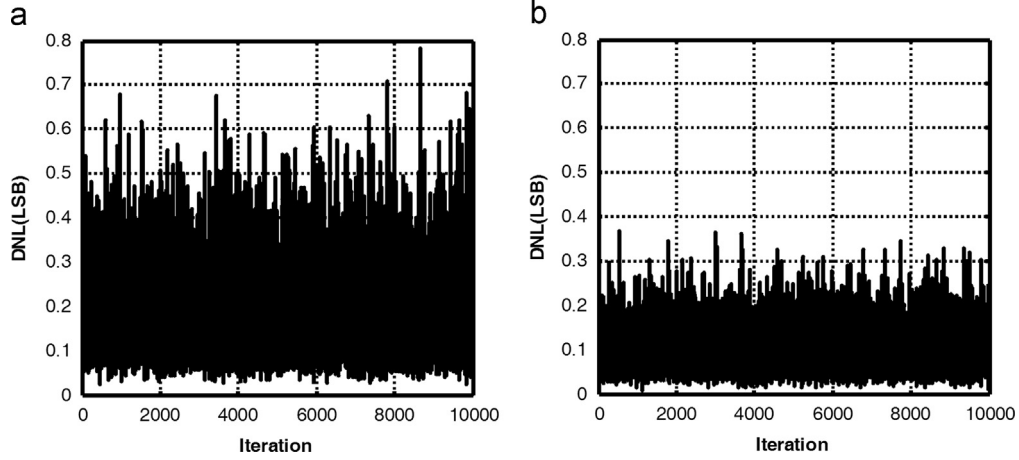


Fig. 4. The comparison of DNL monotonicity performance in 10,000 random iterations between two V_{cm} -based schemes in 10b ADC model with the same unit capacitor. (a) Traditional V_{cm} -based scheme. (b) Proposed split-capacitor V_{cm} -based scheme.

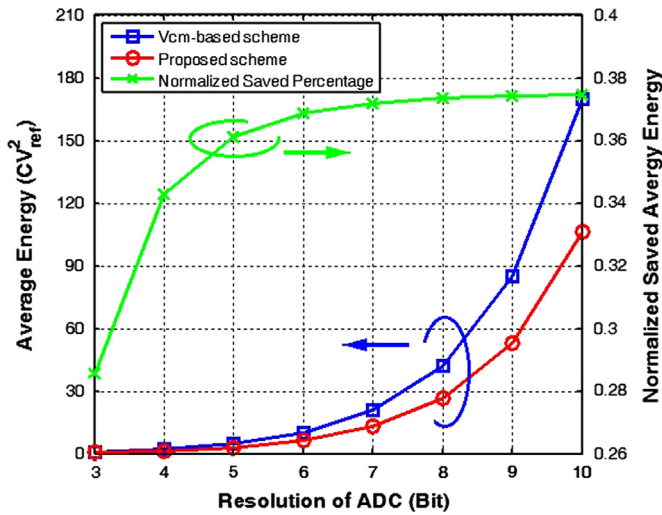


Fig. 5. Average energy consumption at different resolutions.

$$W(1C \text{ in MSB}) = \frac{C}{(2^M - 1)C + C_{PM} + C_{eq,LSB}} = \frac{C(2^L C + C_{PL} + C_{PB})}{(2^{M+L} - 1)C^2 + [2^L C_{PL} + (2^M + 2^L - 2)C_{PB} + 2^L C_{PM}]C + C_{PL}C_{PB} + C_{PL}C_{PM} + C_{PB}C_{PM}} \quad (3)$$

respectively, where

$$C_{eq,MSB} = (C + C_{PB}) / [(2^M - 1)C + C_{PM}] \quad (A // B = AB / (A + B)) \quad (4)$$

$$C_{eq,LSB} = (C + C_{PB}) / [(2^L - 1)C + C_{PL}] \quad (5)$$

C is the unit capacitance, C_{PL} and C_{PM} are the parasitic capacitance on the node LSB_TOP and MSB_TOP and C_{PB} is the parasitic coupling capacitance between the nodes of LSB_TOP and MSB_TOP . To ensure the linearity, the ratio of Eq. (2) over Eq. (3) should equal $1/2^L$, resulting in

$$C_{PL} = (2^L - 1)C_{PB}. \quad (6)$$

If $C_{PL} < (2^L - 1)C_{PB}$, i.e., the weight ratio of 1C in LSBs-array over 1C in MSBs-array is larger than $1/2^L$, a negative periodic DNL characteristic will occur, resulting missing codes eventually. If $C_{PL} > (2^L - 1)C_{PB}$, the unbalance will give rise to a positive periodic DNL characteristic.

Since Eq. (6) is normally not satisfied in this design, an additional capacitance (C_{Added}) is determined by Eq. (7) with the help of the parasitic extraction of C_{PL} , and intentionally added

across the bridge capacitor. By post simulation, the DNL performance of capacitive DAC is controlled within 0.1 LSB.

$$C_{PL} = (2^L - 1)(C_{PB} + C_{Added}) \quad (7)$$

4.2. Comparator circuit with proposed DC offset correction

Fig. 8 shows the comparator which is a two-stage Double-Tail sense amplifier [9] with a proposed DC offset correction cell. During the reset phase when the comparator strobe signal (Latch) is “0”, the middle nodes V_p and V_n of the first stage are charged to the VDD by M4 and M5. Since the current tail M6 is switched off, V_{outp} and V_{outn} are discharged to GND by M7–M8. Once the comparison phase begins at the rising edge of Latch, M4 and M5 stop charging, M3 conducts to ground, and the current tail M6 conducts to power. The charges on V_p and V_n are leak away through the input pair of M1–M2 and M3, which gives rise to the unbalanced V_p and V_n due to the input voltage difference. The second stage amplifies the difference to rail-to-rail swing at the output nodes V_{outp} and V_{outn} [9]. Because of fewer cascade transistors at both stages, this structure is distinguished by the characteristics of high-speed and low voltage operation.

The offset error in the SAR ADC is dominated by the comparator. In this design, the minimum length is adopted for the input transistors for high speed, which, on the other hand, aggravates the comparator offset. Therefore a two-segment DC offset correction scheme is proposed in Fig. 8. The 2-bit MSB segment consists of binary-weighted current-tail arrays to control the middle nodes to discharge differently, extending the correcting range but not affecting the response time. The 3-bit LSB segment consists of capacitor arrays, which has a better linearity performance to meet the accuracy requirement. This scheme can obtain a wide correcting range and sufficient accuracy without increasing the design complexity [7], or degrading the decision speed [10]. As evaluated by simulation, the proposed offset correction scheme achieves about 27 mV correcting range at each side with about 0.47-LSB accuracy.

4.3. Self-timing cell with segmental variable-delay

Fig. 9 shows the variable-delay self-timing cell for the asynchronous SAR ADC [3]. It rationally allocates time slots for the 10-bit steps that compose one determinate conversion cycle. Meanwhile, for hardware simplicity, three delay chains with delay varying from the longest to the shortest selected by Ctrl(5:4) are designed for three segments of bit steps to meet the different

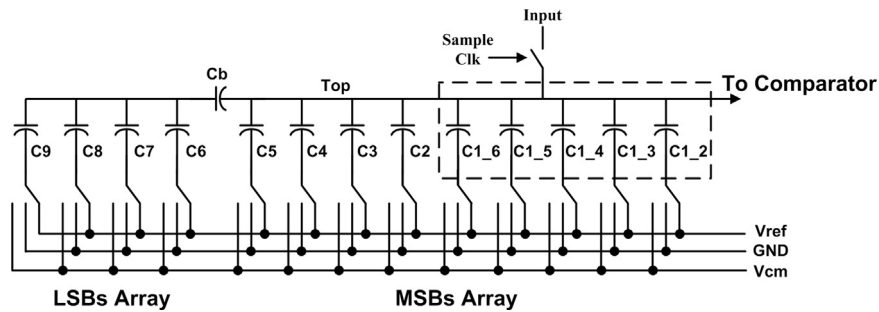


Fig. 6. The actual one-side capacitor-array structure with a bridge structure and the proposed scheme applying at 5-bit MSBs capacitor array.

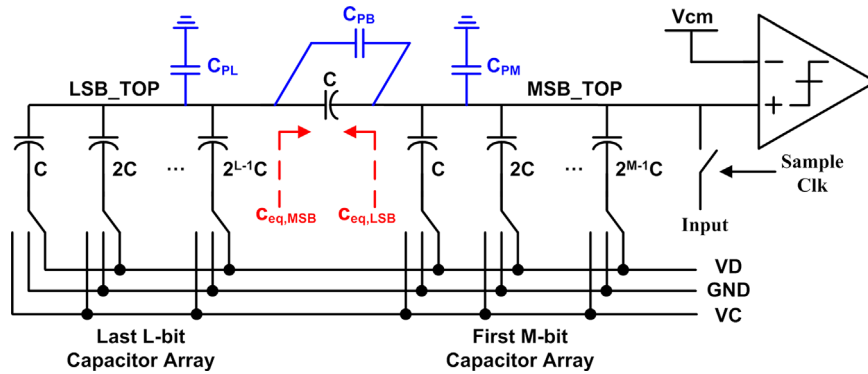


Fig. 7. A general $(M+L+1)$ -bit bridge capacitor-array structure.

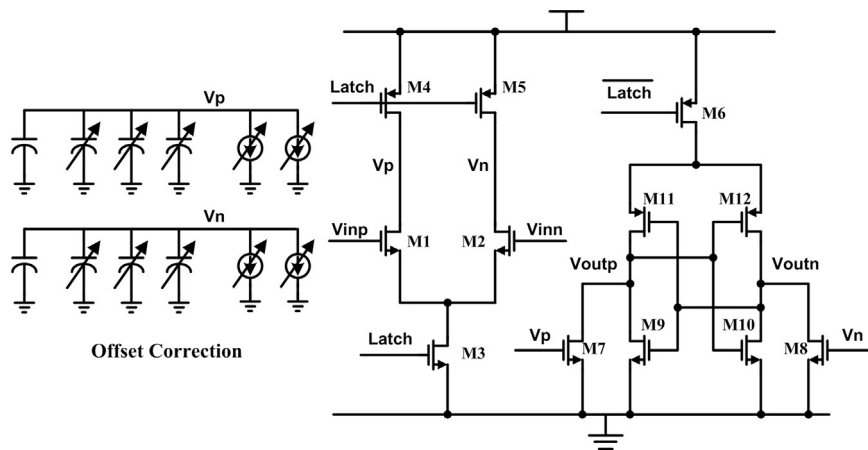


Fig. 8. Comparator with proposed DC offset correction.

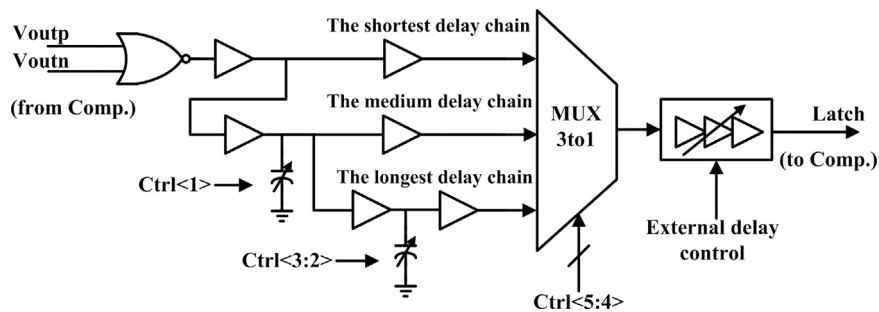


Fig. 9. The self-timing cell with segment-variable delay.

settling time requirements at different bit steps, while adopting Ctrl(3:1) to further regulate the delay at each chain. That is to say, the longest delay chain below is chosen at the 1st and 2nd steps,

the medium delay chain is chosen at the 3rd to 5th steps and the shortest delay chain above is chosen at the 6th to 9th steps, as shown in Fig. 9. An external control signal is also introduced to

tune the delay with a view to the desired settling time to overcome process variations.

4.4. SAR registers with full-custom C^2 MOS DFFs

To flag the on-going bit step in a conversion cycle and save the comparator results of each steps, SAR registers are needed as Fig. 10 shows. It consists of a set of step-flag registers “DFF LIs” and result-saving registers “DFF SIs”. The D-flip-flop (DFF) LIs, the trigger clock of which are the finish flag signal of the comparator Comp.Done, turn to VDD successively to flag the current bit step and then trigger the corresponding DFF SIs to save the comparator result of this bit step. Before a next conversion cycle arrives, all the DFFs will be reset to GND by RST=0.

Both registers are implemented with full-custom C^2 MOS (clocked CMOS) DFF [11], which is more suitable in the low-voltage supply and high-speed applications, compared with the conventional transmission-gate DFF. An inverter-drive feedback pull-up transistor M12 is added at the output node to prevent charge leakage and smooth the final output. Transistors M1, M6 and M11 are used to reset all nodes.

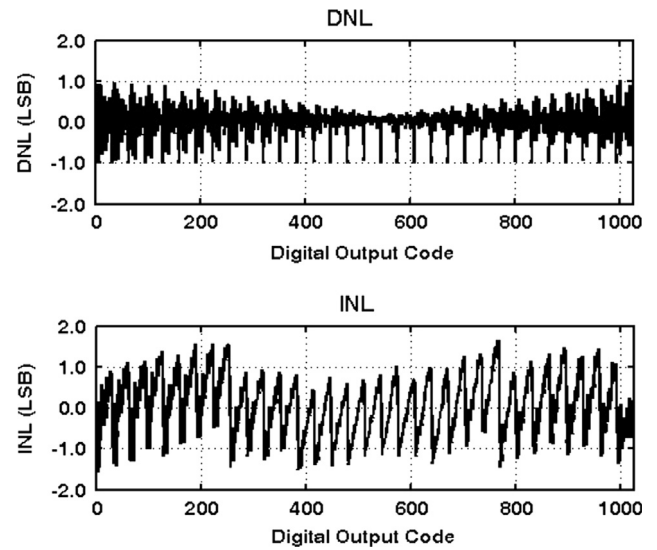


Fig. 12. Measured DNL and INL at 70 MS/s.

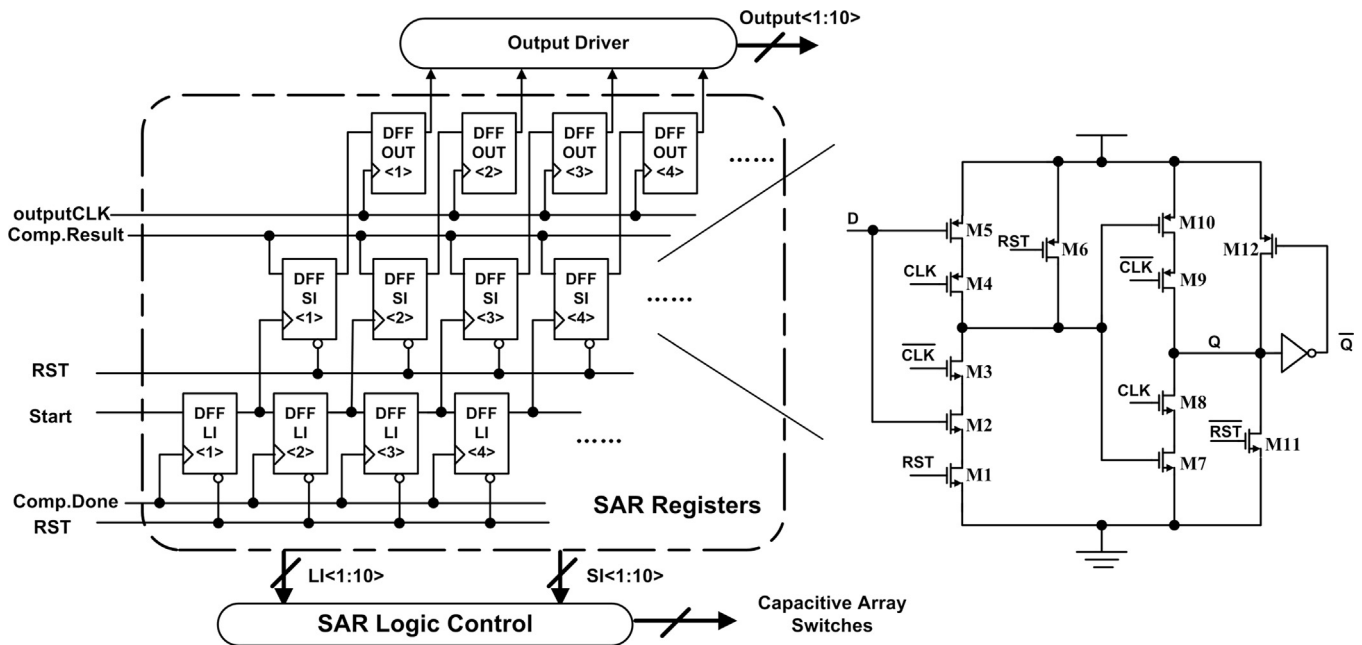


Fig. 10. The structure of SAR registers with full-custom DFFs.

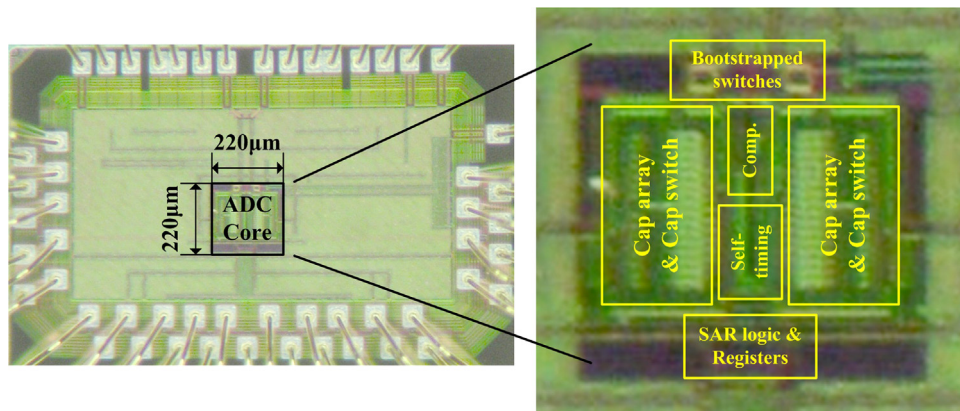


Fig. 11. Chip micrograph and zoomed view.

5. Measurement results and discussion

The prototype ADC was fabricated in TSMC 65-nm 1P9M CMOS technology with metal–insulator–metal (MIM) capacitors. Fig. 11

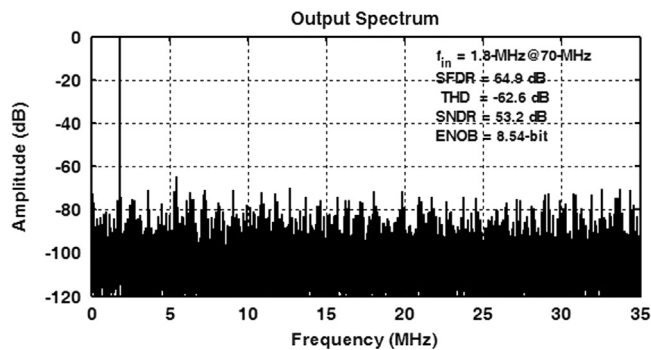


Fig. 13. Measured 65,536-point output spectrum with 1.8 MHz input frequency at 70 MS/s.

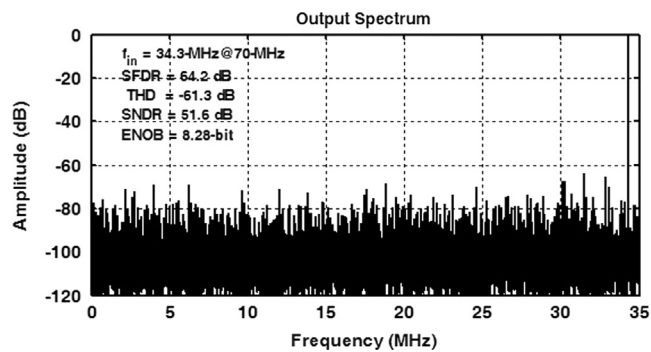


Fig. 14. Measured 65,536-point output spectrum with 34.3 MHz input frequency at 70 MS/s.

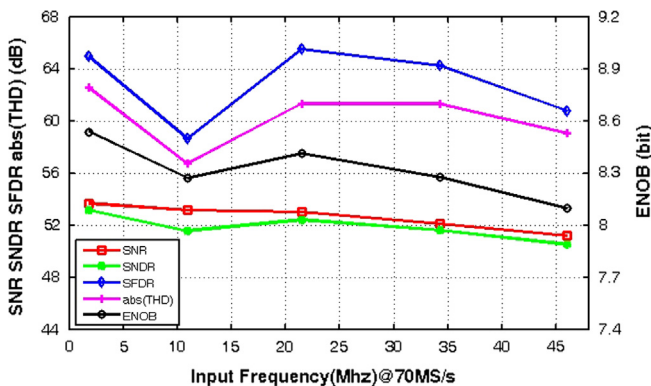


Fig. 15. Dynamic performance versus input frequency at 70 MS/s.

shows the chip micrograph and a zoomed view of the ADC active core which occupies about $220 \times 220 \mu\text{m}^2$. To reduce the high-frequency noise in power supply, large on-chip bypass capacitors are implemented by filling MOS transistors and MOM finger capacitors wherever space is left.

Fig. 12 shows a DNL and INL of 1.01LSB and 1.65LSB, respectively. Figs. 13 and 14 plot the measured output spectrums when the input frequencies are about 1.8 MHz and Nyquist Rate (34.3 MHz), respectively. The dynamic performance versus the input frequency at 70 MS/s is shown in Fig. 15, which indicates the effective resolution bandwidth (ERBW) is 45 MHz. At 1.8 MHz input frequency, the measured SNDR is 53.2 dB. The corresponding effective number of bits (ENOB) is 8.54 bits. When the input frequency is near the Nyquist frequency, the SNDR can still remain 51.6 dB. The total power consumption of the core circuits is $960 \mu\text{W}$ with 1.2 V supply voltage, in which the analog parts and the digital parts consume $780 \mu\text{W}$ and $180 \mu\text{W}$, respectively. According to the power efficiency parameter FoM defined as $\text{Power}/(2^{\text{ENOB}} \times \min\{F_s, 2 \times \text{ERBW}\})$, this ADC achieves an FoM of 36.8 fJ/conversion-step. Table 2 compares this ADC with other state-of-the-art ADCs [1,2,4,5,12,13].

The chip does not achieve the lowest FoM mainly because the bridge-capacitor structure is very sensitive to the parasitic capacitance at the top plates of the LSBs arrays as analyzed in Section 4.1. Although C_{Added} is introduced intentionally by adding MIM capacitor as to compensate the parasitic capacitance C_{PL} on the top plate of the LSBs-array, Eq. (7) can hardly be satisfied in implementation. Firstly, the value of C_{Added} is determined by C_{PL} with the help of parasitic extraction whose accuracy is limited by the parasitic models. Secondly, even if the predetermined C_{Added} value is accurate, it is difficult in implementing any absolute value of capacitance due to the process variation. Owing to these two aspects Eq. (7) is not satisfied. As a result, periodic missing codes occur in the measured DNL performance in Fig. 12 and corresponding harmonics appear in the measured dynamic performance in Figs. 13 and 14. This also reveals that calibration scheme has to be included to further improve the performance of the bridge-capacitor SAR ADC by alleviating its sensitivity to the parasitic effect [7,14].

6. Conclusion

A 10-bit 70-MS/s SAR ADC has been presented in 65-nm CMOS technology. The proposed energy-efficient split-capacitor V_{cm} -based capacitor-switching scheme significantly reduces the switching energy in the capacitive DAC, which is indicated in the theoretical analysis and further verified in this chip. Meanwhile, full-custom SAR logic and registers, two-stage dynamic comparator with proposed DC offset correction scheme and variable-delay self-timing cell are also implemented to ensure the high-speed and accuracy performance. The measurement result shows that it

Table 2
Comparison with the state-of-the-art results.

Specification	[1]	[2]	[4]	[5]	[12]	[13]	This work
Technology (nm)	65	130	90	65	65	65	65
Supply voltage (V)	1.0	1.2	1.2	1.2	1.2	1.1	1.2
Sample rate (MS/s)	50	50	100	100	100	30	70
Resolution (bit)	10	10	10	10	9	10	10
Unit capacitance (fF)	17	4.8	50	3.2	–	63	15
SNDR (dB)	56.9	57.0	56.6	59.0	51.0	56.6	53.2
ENOB (bit)	9.16	9.18	9.1	9.51	8.17	9.11	8.54
Power (mW)	0.82	0.826	3	1.13	1.46	0.85	0.96
FoM (fJ/conv. step)	30	29	77	15.5	39	51	36.8
Active area (mm ²)	0.039	0.052	0.18	0.026	0.012	0.07	0.048

achieves a 70-MS/s operation speed with only 960 μ W power from 1.2 V supply and fulfils 53.2 dB and 51.6 dB SNDR at input rate of 1.8 MHz and Nyquist frequencies, respectively, having a FOM of 36.8 fJ/conversion-step. Discussion also reveals that calibration has to be added for better performance by alleviating the sensitivity of the bridge capacitor to the mismatch of the parasitic capacitance.

Acknowledgments

This project is supported by Doctoral Fund of Ministry of Education of China under Grant 20110071110014 and the State Key Program of National Natural Science of China (Grant no. 61234002).

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